

Synergy of Silver Nanoparticles and Aztreonam against *Pseudomonas aeruginosa* PAO1 Biofilms

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Pathogenic bacterial biofilms, such as those found in the lungs of patients with cystic fibrosis (CF), exhibit increased antimicrobial resistance, due in part to the inherent architecture of the biofilm community. The protection provided by the biofilm limits antimicrobial dispersion and penetration and reduces the efficacy of antibiotics that normally inhibit planktonic cell growth. Thus, alternative antimicrobial strategies are required to combat persistent infections. The antimicrobial properties of silver have been known for decades, but silver and silver-containing compounds have recently seen renewed interest as antimicrobial agents for treating bacterial infections. The goal of this study was to assess the efficacy of citrate-capped silver nanoparticles (AgNPs) of various sizes, alone and in combination with the monobactam antibiotic aztreonam, to inhibit *Pseudomonas aeruginosa* PAO1 biofilms. Among the different sizes of AgNPs examined, 10-nm nanoparticles were most effective in inhibiting the recovery of *P. aeruginosa* biofilm cultures and showed synergy of inhibition when combined with sub-MIC levels of aztreonam. Visualization of biofilms treated with combinations of 10-nm AgNPs and aztreonam indicated that the synergistic bactericidal effects are likely caused by better penetration of the small AgNPs into the biofilm matrix, which enhances the deleterious effects of aztreonam against the cell envelope of *P. aeruginosa* within the biofilms. These data suggest that small AgNPs synergistically enhance the antimicrobial effects of aztreonam against *P. aeruginosa* *in vitro*, and they reveal a potential role for combinations of small AgNPs and antibiotics in treating patients with chronic infections.

Bacterial biofilms are communities of cells that are attached to abiotic or biotic surfaces and are encased in an extracellular matrix consisting of secreted proteins, polysaccharides, nucleic acids, and cellular debris (1, 2). Biofilms are linked to increased antimicrobial resistance in chronic infections, such as those caused by the opportunistic Gram-negative pathogen *Pseudomonas aeruginosa* in compromised individuals suffering from burns, cancer, or AIDS or in the lungs of patients with cystic fibrosis (CF) (1, 3, 4). Increased resistance, due in part to the inherent protective architecture of the biofilm, limits the absorption of antibiotics to target sites on and within the bacterial cells and increases the MICs of antimicrobial agents 100- to 1,000-fold (5). With the adaptability and metabolic versatility of this organism (6, 7), biofilm-mediated chronic *P. aeruginosa* infections represent the major factor leading to the increased morbidity rates and premature death seen in patients with CF (8, 9). This makes *P. aeruginosa* an important model organism for study of bacterial biofilm development, antibiotic resistance, and antibiofilm treatments (10, 11).

The rise of multidrug-resistant bacteria and the difficulty of treating chronic biofilm-mediated infections (3, 10, 12) have prompted renewed interest in using silver-containing compounds to treat these infections (13, 14). The antimicrobial effects of silver and silver-containing compounds have been known for decades (13–15); however, despite their historical use and recent resurgence in consumer products, little is known about their precise mode of antimicrobial action (15–20). Recent studies have revealed the antimicrobial effects of silver ions and silver nanoparticles (AgNPs) in combination with commonly used antibiotics (21–24). These findings, although largely derived from use against planktonic bacterial cultures, highlight the finding that antimicrobial synergies exist when silver is combined with traditional treatments, with the combined effects being greater than the sum of the

individual effects. Moreover, these synergies may be particularly effective in the treatment of biofilm-mediated infections.

In this study, we examined the efficacy of combining citrate-capped AgNPs of different sizes with the monobactam antibiotic aztreonam to prevent *P. aeruginosa* PAO1 growth and biofilm recovery *in vitro*. Aztreonam is a relatively recent addition to the antibiotic treatments utilized for CF patients (25). Thus, it is of interest to evaluate effective combination treatments to aid in minimizing the potential for the development of bacterial resistance. Aztreonam in combination with tobramycin has been evaluated against *P. aeruginosa* PAO1 biofilms in a coculture model using CF-derived airway cells (26). Several CF-derived *P. aeruginosa* strains did show aztreonam tolerance (26); thus, additional research to evaluate other combination therapies to enhance the efficacy of aztreonam would benefit its use for CF patients. We hypothesized that combining AgNPs and aztreonam would enhance their activities against preformed *P. aeruginosa* PAO1 biofilms, compared with each antimicrobial alone. Our results demonstrated that smaller AgNPs were more effective, alone and in combination with aztreonam, in reducing *P. aeruginosa* PAO1 biofilm biomass and viability than were larger nanoparticles. Smaller AgNPs displayed synergy in preventing biofilm recovery when combined with aztreonam. Transmission electron microscopy (TEM)

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and full-field optical coherence tomography (FFOCT) revealed that this synergy was enhanced by the ability of smaller AgNPs to penetrate into the biofilms, and this activity directly affected biofilm architecture. Together, these findings highlight the potential for AgNPs to enhance the activity of commonly used antibiotics against chronic *P. aeruginosa* biofilm-mediated infections.

MATERIALS AND METHODS

Antibiotic, silver nanoparticles, and chemicals. Aztreonam and citrate-capped silver AgNPs of various sizes (10, 20, 40, 60, and 100 nm) were purchased from Sigma-Aldrich (St. Louis, MO). Minimal Davis broth without dextrose (Difco) was purchased from Thermo Fisher (Mississauga, Ontario, Canada). The minimal Davis broth was prepared and autoclaved according to the manufacturer's instructions and then was supplemented with filter-sterilized dextrose (10% [wt/vol]) to a final concentration of 0.1% (wt/vol). Minimal Davis medium supplemented with dextrose (MDM) was selected for the biofilm assays to provide a simpler, more-defined system to evaluate the effects of aztreonam and the AgNPs against *P. aeruginosa* biofilms, in comparison with some of the more nutrient-rich media (e.g., cation-adjusted Mueller-Hinton broth or tryptic soy broth) typically used for such assays (27). The aztreonam stock solution was prepared in MDM at a concentration 4 times higher than the highest antibiotic concentration used in the minimum biofilm eradication concentration (MBEC) assays (i.e., 2,048 µg/ml), and these stock solutions were stored at −20°C.

Bacterial strain. *Pseudomonas aeruginosa* PAO1 (28) was the model microorganism utilized for these studies. Cultivation of this microorganism for the MBEC and silicone tube biofilm assays was performed according to the MBEC assay manufacturer's instructions.

Minimum biofilm eradication concentration assays. The MBEC assays were set up and performed as described previously (16) and involved growth, challenge, and recovery phases. All tests were performed in triplicate, using independent PAO1 cultures, and all manipulations of the MBEC assay plates were performed in a biological safety cabinet. For the growth phase, the MBEC assay plate was placed in a humidified container following inoculation, placed in an incubator at 37°C, and shaken at 100 rpm for 20 h to develop the biofilms. Following the incubation, the optical density at 600 nm (OD_{600}) of the planktonic cultures was determined. The peg biofilms were rinsed and placed in microtiter plates containing different combinations of aztreonam and AgNPs, prepared using a microdilution two-dimensional (2D) checkerboard procedure (23). The setup of each challenge plate evaluated 2-fold dilutions of aztreonam (highest concentration, 512 µg/ml; lowest concentration, 2.00 µg/ml) with 2-fold dilutions of AgNPs of a particular size (highest concentration, 10.0 µg/ml; lowest concentration, 0.156 µg/ml). To determine the MBEC of aztreonam alone, a separate test was conducted evaluating aztreonam concentrations between 500 and 4,000 µg/ml. Within each test, MDM only (negative growth control) and MDM only with biofilms (untreated; positive growth control) were included. The challenge plate was incubated as described for the growth phase. Following 20 h of incubation, peg biofilms were rinsed, placed in a microtiter plate containing 200 µl MDM/well, and incubated as described previously. Following incubation, biofilm viability, MBEC values, and biofilm biomass were determined, as described below.

Biofilm viability and MBEC determination. Following the 20 h of incubation of the recovery phase, the OD_{600} of the planktonic cultures in the wells of the microtiter plate were measured. The OD_{600} values were utilized as surrogate measures for biofilm viability, instead of determining CFU/ml values for each well. This was to accommodate the very large number of samples that were analyzed, which made determining CFU/ml values prohibitive. Also, the specific determination of a CFU/ml value for each well was not critical; rather, it is the reestablishment of a planktonic culture that can be used as a qualitative indicator that a particular AgNP-aztreonam combination was not successful in eliminating or inhibiting the biofilms. Biofilm viability was normalized and expressed as a percent-

age of untreated control values. The OD_{600} values were also used for MBEC determinations (29, 30).

Biofilm biomass determination. Following the 20 h of incubation of the recovery phase, the lid was placed in a microtiter plate with 200 µl phosphate-buffered saline (PBS) in each well, to rinse the pegs of any loosely adherent cells for 1 to 2 min. The biofilms were then processed and stained with crystal violet to measure biomass (31, 32). After rinsing of the peg lid, the biofilms were fixed in 99% methanol for 15 min and then air dried. The peg biofilms were then stained with 0.2% crystal violet (wt/vol) for 5 min. The biofilms were rinsed twice in water to remove excess stain and were allowed to air dry. To solubilize the bound dye, the stained biofilms were placed in 33% glacial acetic acid for 10 min, to ensure that all of the dye was solubilized. The absorbance of each well was measured at 590 nm. These values were normalized and expressed as percentages of untreated control values.

Silicone tube assays. Biofilms were grown in a once-through biofilm tubing reactor (4, 33, 34); the setup, biofilm formation, and antimicrobial challenge were performed at 37°C. A 10-liter carboy containing sterile MDM was supplied through the tubing by a peristaltic pump. Approximately 60 cm of silicone tubing (Cole Parmer Masterflex size 14; inner diameter, 1.6 mm) was used, with a central 20-cm section that was cut and reattached with plastic connectors to define an area of tubing for inoculation and challenge with antimicrobials (volume, 0.402 cm³). All tubing and media were autoclaved prior to use. Prior to inoculation, the entire length of tubing was preconditioned with MDM for 15 min, at a flow rate of ~0.2 ml/min. To inoculate the tubing, the flow was stopped and the PAO1 culture (2 ml, prepared as described above) was injected by needle (at ~1 ml/min) into the 20-cm section, immediately upstream of the plastic connector. This was followed by no flow for 1 h, to allow initial microbial attachment to occur. After this period, the flow of MDM was restarted and the biofilms were developed for 72 h.

Following biofilm development, the flow was stopped and the biofilms were treated with 64.0 µg/ml aztreonam and/or 0.625 µg/ml AgNPs (either 10 nm or 100 nm in size). An untreated growth control was also prepared with 2 ml of MDM alone. Treatments were injected by needle (at ~1 ml/min) into the 20-cm section, immediately upstream of the plastic connector. This volume was sufficient to flush the entire 20-cm segment of planktonic cells and to provide consistent amounts of aztreonam and/or AgNPs throughout the segment. The tubing was incubated statically for up to 18 h. Sections of tubing (length, 1 cm) were cut from the 20-cm segment at defined time points posttreatment (0, 8, 12, 18, and 24 h), to evaluate any changes in biofilm thickness (via full-field optical coherence tomography [FFOCT]) and cell morphology (via transmission electron microscopy [TEM]). The tube sections were placed in 1.5-ml microcentrifuge tubes with 2.5% glutaraldehyde in 0.1 M sodium cacodylate buffer to fix the samples, and the samples were stored at 4°C until analysis. Based on our initial observations, the 18-h samples were selected for evaluation with FFOCT and TEM.

Full-field optical coherence tomography and transmission electron microscopy. An FFOCT system (LightCT scanner; LLTech SAS, Paris, France) was used to evaluate biofilm thickness on the luminal surface of the fixed silicone tube samples. All samples were scanned using the same parameters as described below. The native field defined by the FFOCT system provides a tangential image of 0.8 mm by 0.8 mm, with transverse and lateral resolutions of 1.5 µm and 1 µm, respectively. A fixed silicone tube section was placed in the sample holder with the lumen side up. To scan the entire luminal area of the silicone tube opening, with a 1.6-mm diameter, a grid of 2.4 mm by 2.4 mm was produced by stitching 9 neighboring native fields in the *x-y* plane (3-by-3 grid). Five hundred stacks of these fields were acquired to evaluate the biofilm present on the silicone tube surface to a depth of 500 µm (a volume of ~1 mm³ was scanned). One hundred acquisitions (the maximum) were averaged for each image, each complete scan lasted ~5 h, and 4,500 native fields were captured. Following scanning, the tube sections were removed from the sample

holder, placed back in their respective microcentrifuge tubes, and stored at 4°C until they were processed for TEM.

The 500 *en face* two-dimensional (2D) images acquired with the FFOCT system were exported and visualized with medical-grade viewing software. A measurement tool included as part of the software was used to determine the biofilm thickness at ~75-μm intervals across the entire length measured, and six locations were analyzed per tube section. For each *en face* 2D image, 10 measurements of the biofilm thickness were taken across the entire luminal wall. In total, 60 measurements were taken per sample and averaged. The same software was used to generate three-dimensional (3D) reconstructions of the biofilms on the tubing surface.

Biofilm samples prepared for TEM were rinsed twice with 0.1 M phosphate buffer (pH 7.4) for 5 min and fixed for 0.5 h with 1% (wt/vol) osmium tetroxide in 0.1 M phosphate buffer (pH 7.4). Samples were washed twice with nanopure water (5 min) and subjected to an ethanol dehydration series, embedding in LR White embedding medium, and thin sectioning as described previously (35). Grids containing thin sections were negatively stained for 7 min with 2% uranyl acetate, washed with high-performance liquid chromatography (HPLC)-grade water, and then stained with Reynold's lead citrate for 3 min prior to viewing. Samples were viewed using an FEI Technai G2 F20 transmission electron microscope, operating at 200 kV and equipped with a bottom-mount Gatan 4k charge-coupled-device (CCD) camera, under standard operating conditions.

Bliss model of synergy and fractional inhibitory concentration index calculations. Aztreonam-AgNP synergism was calculated using the Bliss independence model (35–38) and the fractional inhibitory concentration (FIC) index (39, 40) to determine the nature of the inhibitory effects observed when different antimicrobials were combined. The FIC index was calculated as described previously in a number of studies (39, 40). Briefly, the FIC index was calculated for the first nonturbid well for each row and column along the turbidity/nonturbidity interface, representing the AgNP-aztreonam combinations that inhibited the *P. aeruginosa* PAO1 biofilm, i.e., MBEC. This was determined by examining planktonic growth during the recovery step following biofilm challenge with aztreonam and/or AgNPs. OD₅₉₀ values of ≤0.01 were considered to indicate nonturbidity. We used the equation $\Sigma FIC = FIC_A + FIC_B = (C_A/MBEC_A) + (C_B/MBEC_B)$, where MBEC_A and MBEC_B are the MBECs of AgNPs and aztreonam alone, respectively, and C_A and C_B are the concentrations of AgNPs and aztreonam in combination, respectively. The FIC index median and range were determined. As is standard, FIC index values of ≤0.5 were considered to indicate synergism, values of >0.5 and <4 no interaction/additivity, and values of ≥4 antagonism.

For the Bliss model, we used the formula $S = (f_{X0}/f_{00})(f_{0Y}/f_{00}) - (f_{XY}/f_{00})$, where f_{XY} refers to the *P. aeruginosa* PAO1 biofilm biomass in the presence of the combined antimicrobials at concentration X for aztreonam and concentration Y for AgNPs, f_{X0} and f_{0Y} refer to the biofilm biomass in the presence of the individual antimicrobials at concentrations of X and Y, respectively, f_{00} refers to the biofilm biomass in the absence of antimicrobials, and S corresponds to the degree of synergy. Positive values for S reflect a degree of synergy between aztreonam and AgNPs, while a negative value for S reflects an antagonistic interaction (35–38). The calculated degree of antimicrobial synergy (Bliss coefficient) was determined for all of the different AgNP-aztreonam combinations. However, we present representative data for all concentrations of AgNPs with high (512 μg/ml), moderate (64.0 μg/ml), and low (4.00 μg/ml) aztreonam concentrations, against *P. aeruginosa* biofilms. These data were used to determine the combinations and concentrations of AgNPs and aztreonam to be used in the silicone rubber tubing assays. For the 10-nm AgNP tests, the 64-μg/ml aztreonam concentration paired with the 0.625-μg/ml AgNP concentration provided a test to evaluate a synergistic interaction, while the same concentrations for the 100-nm AgNPs and aztreonam suggested no synergistic interaction. These combinations were compared in the silicone rubber tubing assay.

Statistical analysis. Statistical analyses of the data were completed using Prism 5 (GraphPad Software, San Diego, CA). FFOCT measurements were analyzed using one-way analysis of variance (ANOVA) with a Tukey-Kramer multiple-comparison *post hoc* test.

RESULTS

High-throughput biofilm antimicrobial susceptibility assay. To assess the antimicrobial properties of citrate-capped, dispersed AgNPs of different sizes (Fig. 1A), alone and in combination with the antibiotic aztreonam, we used the minimum biofilm eradication concentration (MBEC) high-throughput assay (29, 30, 41), which has been determined to reflect antimicrobial efficacy against biofilms more accurately than MIC assays (16). These assays involve specialized MBEC plate lids with pegs that support surface-attached bacterial growth (i.e., biofilms) and wells that support free-living (i.e., planktonic) populations. Sequential transfers of the plate lids containing the biofilms to different wells allow for the analysis to be separated into three distinct stages, i.e., growth, challenge, and recovery (Fig. 1B). This experimental design also permits determination of the MIC for planktonic cells and the MBEC within the same assay (42–44). The planktonic population of bacteria in the wells after the challenge-phase incubation (i.e., in the presence of antimicrobial) reflects the planktonic susceptibility of cells shed from the biofilm (42, 44) and is used to calculate the MIC; the planktonic population in the wells after the recovery-phase incubation (i.e., in fresh medium) indicates postchallenge biofilm viability (29, 30, 42, 43) and is used to calculate the MBEC (29, 30). Following the recovery incubation, the biofilm-covered pegs are stained with crystal violet to determine the biofilm biomass.

The reproducibility of biofilm formation in the MBEC system during the growth phase was measured as the ratio of planktonic growth to the corresponding biofilm biomass accumulation in three independent growth experiments. Our analysis demonstrated consistent and reproducible biofilm biomass accumulation and planktonic growth using a standard starting inoculum of *P. aeruginosa* PAO1 (Fig. 1C). The values of the mean ratio (± standard error of the mean) of planktonic growth to biofilm formation were 0.198 ± 0.002 , 0.183 ± 0.002 , and 0.192 ± 0.001 for experiments 1, 2, and 3, respectively. Thus, although some variations occurred between microtiter plates, as expected, variations within plates were negligible. To permit analysis across experiments, the data were normalized to values for untreated controls, as described in Materials and Methods.

Small AgNPs and aztreonam individually inhibit planktonic cell growth during antimicrobial challenge. After the growth phase, preformed biofilms were incubated in a second 96-well microtiter plate, containing different concentrations and combinations of AgNPs and/or aztreonam (challenge) (Fig. 1B). We first determined MIC values for the planktonic populations shed from the biofilms in the presence of various concentrations of either AgNPs or aztreonam alone (Table 1). The median and mode MIC values demonstrated that, when administered individually, smaller AgNPs were more effective at inhibiting *P. aeruginosa* planktonic growth in the challenge phase than were larger AgNPs (Table 1). Median MICs for AgNPs alone ranged from 0.234 μg/ml for the 10-nm particles to 7.50 μg/ml for the 100-nm particles. For all concentrations of aztreonam tested (2.00 μg/ml to 512 μg/ml), no planktonic growth was observed in the presence of antibiotic. These results indicated MIC values of ≤2.00 μg/ml for

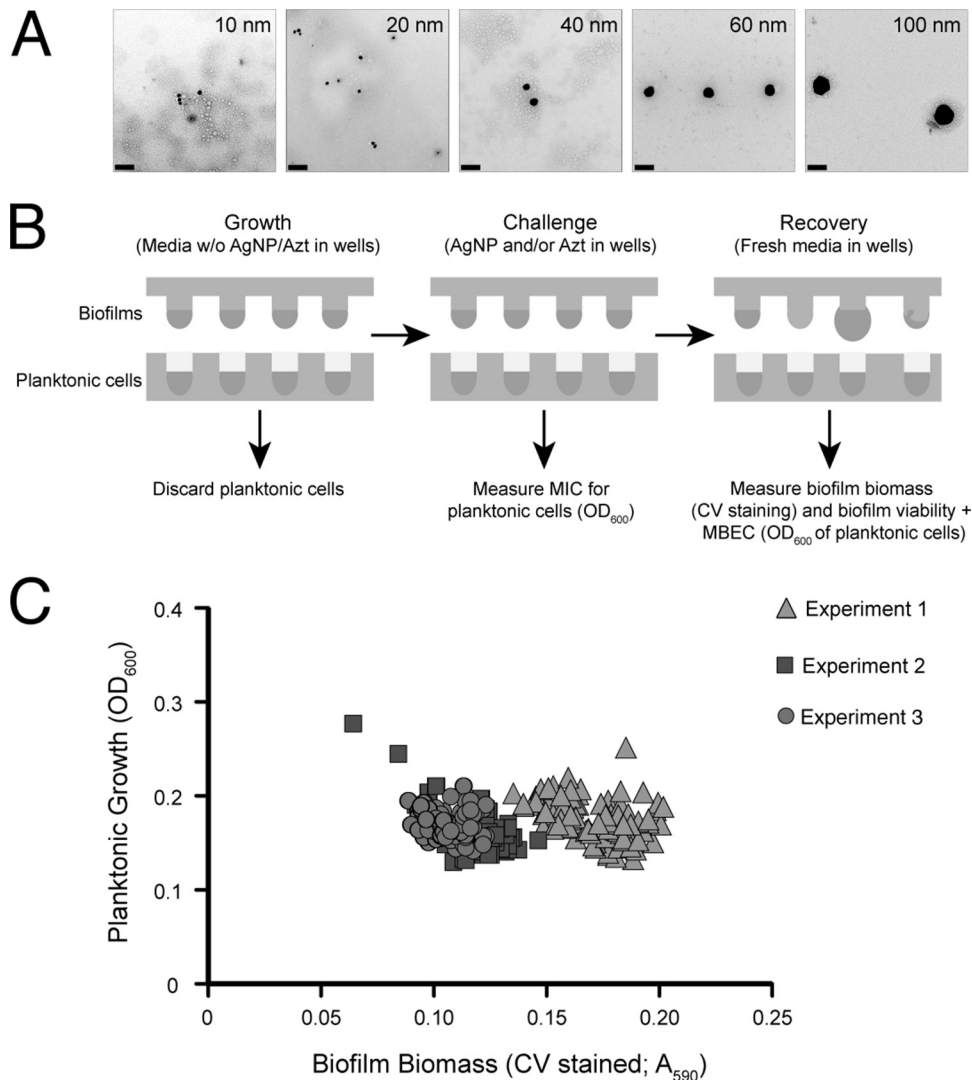


FIG 1 High-throughput MBEC biofilm antimicrobial susceptibility assay. (A) Transmission electron micrographs of AgNPs (10 nm to 100 nm) used in this study. Bars, 100 nm. (B) Schematic illustration of the workflow used (i) to culture *P. aeruginosa* PAO1 planktonic and biofilm cultures (growth phase), (ii) to incubate biofilms in the presence of AgNP-aztreonam treatments and to determine MIC values for the planktonic cultures (challenge phase), and (iii) to determine biofilm biomass, viability, and MBEC values following incubation in fresh medium without antimicrobials (recovery phase) (see Materials and Methods for more details). (C) Reproducibility of biofilm formation, assessed in the growth phase of the experiment by plotting planktonic growth (based on planktonic cell density in the wells of the growth phase) versus biofilm growth (based on accumulated bacterial cell density, stained with crystal violet [CV], in biofilms in the growth phase) ($n = 288$). A_{590} , absorbance at 590 nm.

TABLE 1 MICs observed for individual antimicrobials against planktonic cells following 20-h challenge of *P. aeruginosa* PAO1 biofilms

Treatment	MIC ($\mu\text{g/ml}$)			<i>n</i>
	Range	Median	Mode	
10-nm AgNPs	0.156–0.625	0.234	0.156	8
20-nm AgNPs	0.312–2.50	0.625	1.25	8
40-nm AgNPs	0.312–2.50	1.25	1.25	8
60-nm AgNPs	1.25–10.0	2.50	2.50	7
100-nm AgNPs	2.50–10.0	7.50	10.0	6
Aztreonam ^a	ND	≤ 2.00	≤ 2.00	37

^a For all concentrations of aztreonam tested (2.00 $\mu\text{g/ml}$ to 512 $\mu\text{g/ml}$), no planktonic growth was observed following challenge. Aztreonam concentrations lower than 2.00 $\mu\text{g/ml}$ were not tested, since antibiotic ranges were selected based on investigation of the inhibition of biofilm recovery after the challenge phase. The median MIC of aztreonam for planktonic cells in the challenge phase of this experiment was therefore determined to be ≤ 2.00 $\mu\text{g/ml}$. ND, not determined.

planktonic cells shed from biofilms treated with aztreonam in the challenge phase in this system (Table 1). Lower concentrations of aztreonam were not tested here, since our focus was on investigating inhibition of biofilm growth using these antimicrobials, as discussed below.

Aztreonam treatment alone enhances biofilm biomass accumulation in the recovery phase. After the challenge phase, treated biofilms were rinsed, moved to a third 96-well microtiter plate containing fresh medium, and incubated for 20 h to determine biofilm biomass and viability (recovery) (Fig. 1B). Biofilms treated with increasing concentrations of aztreonam alone during the challenge phase demonstrated significant increases in biomass after the 20-h recovery, with the highest concentrations of antibiotic leading to an average of $\sim 250\%$ recovered biomass, compared to untreated biofilms (Fig. 2A). Biofilm viability was similar

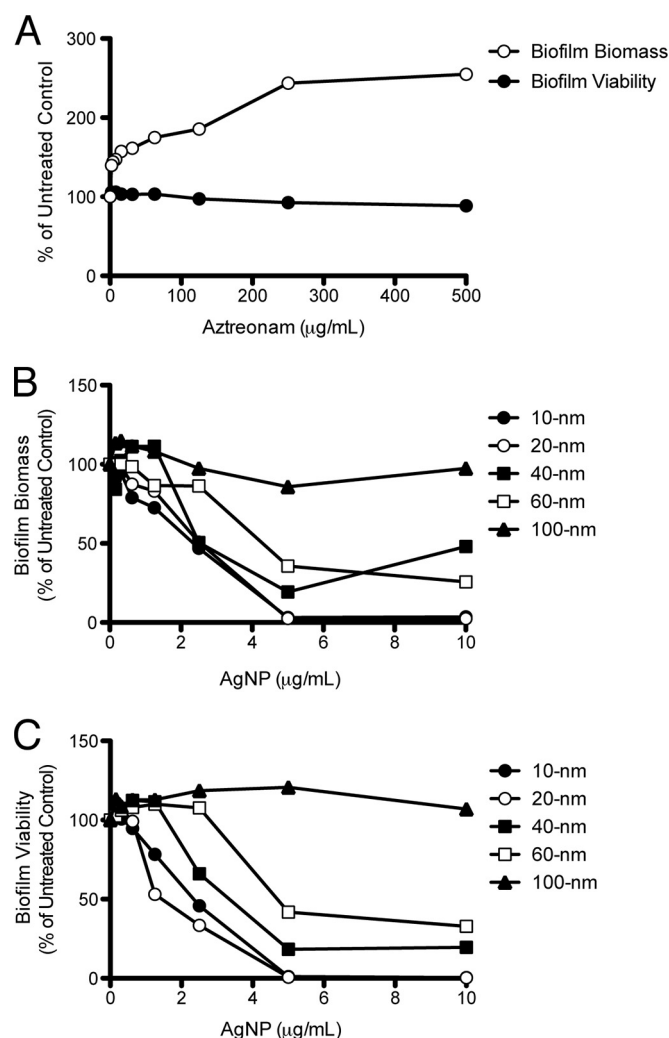


FIG 2 Effects of AgNPs or aztreonam alone on *P. aeruginosa* PAO1 biofilm biomass and viability. Values were normalized to those obtained from untreated control samples after biofilms were incubated in fresh medium in the recovery phase. (A) Biofilm biomass and viability following challenge with aztreonam (0.0 µg/ml to 512 µg/ml). (B and C) Biofilm biomass (B) and viability (C) following challenge with different sizes (10 nm to 100 nm) and concentrations (0.0 µg/ml to 10.0 µg/ml) of AgNPs.

to that of untreated biofilms (~100% with all concentrations tested) (Fig. 2A). The MBECs for *P. aeruginosa* biofilms following challenge with aztreonam alone were also evaluated (Table 2). These data suggest that the inhibitory effects of this antibiotic alone against preformed biofilms are limited and potentially enhance biofilm growth after antibiotic treatment has ceased.

Small AgNPs alone are more effective than larger nanoparticles in reducing biofilm biomass and viability. Treatment of *P. aeruginosa* PAO1 biofilms with AgNPs alone revealed that smaller AgNPs were more effective than larger nanoparticles in limiting or reducing biofilm biomass in the recovery phase, compared to untreated biofilms (Fig. 2B). Both 10-nm and 20-nm AgNPs effectively reduced biofilm biomass at concentrations of ≥ 0.625 µg/ml, while the 40-nm and 60-nm AgNPs reduced biofilm biomass at concentrations of ≥ 2.50 µg/ml. The 100-nm AgNPs demonstrated negligible effects on *P. aeruginosa* biofilm biomass, result-

ing in ~100% biomass recovery at all concentrations tested, similar to untreated samples (Fig. 2B). Similar postchallenge trends were observed for biofilm viability (Fig. 2C). The 10-nm and 20-nm AgNPs showed the most significant reductions in biofilm viability after the challenge, with reduced viability occurring at concentrations of ≥ 1.25 µg/ml. The 40-nm AgNPs reduced biofilm viability at ≥ 2.50 µg/ml, while the 60-nm AgNPs reduced viability at ≥ 5.00 µg/ml. Biofilm viability following treatment with 100-nm AgNPs was consistently greater than 100% at all concentrations tested (Fig. 2C), suggesting that the larger size of these nanoparticles reduced their effectiveness as antibiofilm antimicrobials. We also calculated MBEC values for the AgNPs of various sizes (Table 2). Within the parameters tested, the largest changes in MBEC values, relative to MIC values (Table 1), were observed for the 10-nm AgNPs; the median and mode MBEC values were 16- and 32-fold higher than the corresponding MIC values, respectively.

Specific combinations of AgNPs and aztreonam inhibit biofilm biomass and viability. Next, we evaluated the recovery of *P. aeruginosa* biofilms after challenge with different combinations of AgNPs and aztreonam (Fig. 3). In the absence of AgNPs, the biofilm biomass was dramatically increased as the concentration of aztreonam used in the challenge phase was increased (Fig. 3A). For the 10-nm AgNPs, the first observations of significant reductions of biofilm biomass occurred at a concentration of 0.312 µg/ml, with various levels of reduction depending on the concentration of aztreonam (Fig. 3C). Increasing the 10-nm AgNP concentration to 0.625 µg/ml dramatically reduced the biofilm biomass at aztreonam concentrations of ≥ 8.00 µg/ml (Fig. 3D). As the AgNP concentrations were increased to 2.50 µg/ml (Fig. 3E and F), biofilms treated with 20-nm AgNPs displayed greatly reduced biomass, similar to those treated with 10-nm AgNPs (~2% of values for untreated controls), those treated with 40-nm AgNPs displayed biomass reduced to less than ~50%, and those treated with 60-nm AgNPs were similar to untreated cells (~100%). This trend appeared to continue up to the highest concentration of AgNPs tested (10 µg/ml), at which biomass values for the largest nanoparticles began to fall below 100% (Fig. 3H).

The viability of *P. aeruginosa* biofilms after challenge with different combinations of AgNPs and aztreonam followed a similar pattern overall, compared to the biomass results, with the excep-

TABLE 2 Minimum biofilm eradication concentrations determined for individual antimicrobials by measuring the inhibition of biofilm viability following 20 h of recovery of *P. aeruginosa* PAO1 biofilms^a

Treatment	MBEC (µg/ml)			n
	Range	Median	Mode	
10-nm AgNPs	1.25–5.00	3.75	5.00	6
20-nm AgNPs	1.25–5.00	3.75	5.00	6
40-nm AgNPs	2.50 to >10.0 ^b	>2.50 ^b	>5.00 ^b	6
60-nm AgNPs	5.00 to >10.0 ^b	>5.00 ^b	>10.0 ^b	6
100-nm AgNPs	ND ^c	>10.0 ^c	>10.0 ^c	5
Aztreonam	4,000	4,000	4,000	9

^a Biofilm viability was assessed by measuring the OD₆₀₀ of the planktonic cultures in the wells of the microtiter plate following the 20 h of incubation of the recovery phase.

^b Biofilms remained viable (i.e., planktonic growth was observed) following recovery at the highest concentration of AgNPs tested (10.0 µg/ml).

^c Biofilms remained viable (i.e., planktonic growth was observed) following recovery for all concentrations of AgNPs tested (0.156 µg/ml to 10.0 µg/ml). ND, not determined.

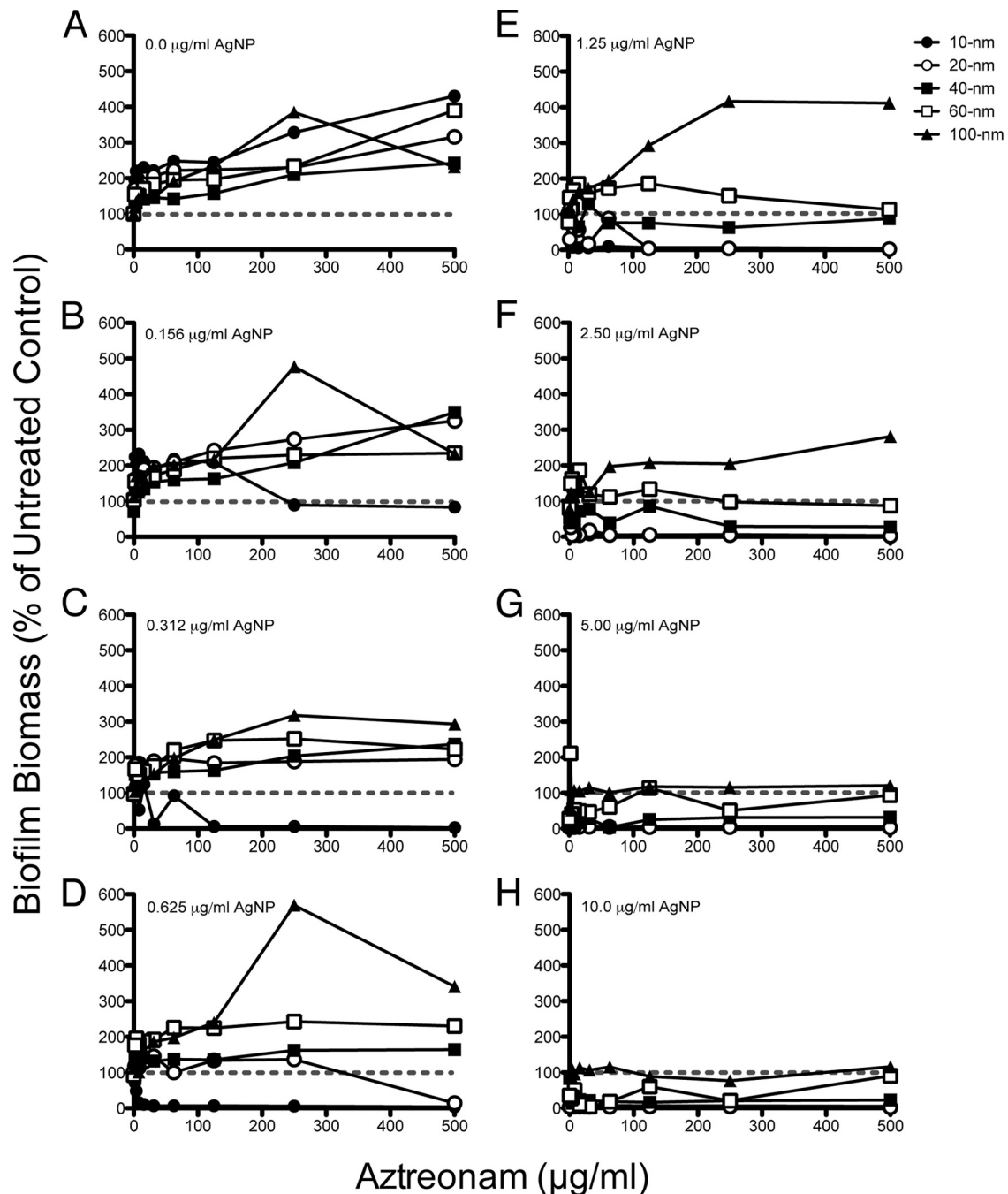


FIG 3 Effects of different AgNP-aztreonam combinations on *P. aeruginosa* PAO1 biofilm biomass in the recovery phase. Normalized values of percent recovery of biofilm biomass following challenge with different combinations of aztreonam (0.0 $\mu\text{g/ml}$ to 512 $\mu\text{g/ml}$) and AgNP sizes (10 nm to 100 nm) at the specific AgNP concentration (0.0 $\mu\text{g/ml}$ to 10.0 $\mu\text{g/ml}$) indicated in the top left corner of each panel are shown (panel A represents replicate samples treated with aztreonam only). Biofilm biomass values for treated samples were normalized to values for untreated cells (i.e., 0.0 $\mu\text{g/ml}$ AgNPs and 0.0 $\mu\text{g/ml}$ aztreonam). Dashed lines, 100% recovery (i.e., levels observed for untreated cells).

tion of the 100-nm particles (Fig. 4). In the absence of AgNPs, biofilm viability after treatment with increasing concentrations of aztreonam recovered to $\sim 100\%$ of untreated controls (Fig. 4A). At the lowest AgNP concentration of 0.156 $\mu\text{g/ml}$, the 10-nm AgNPs significantly decreased biofilm viability (from $\sim 100\%$ to $\sim 60\%$) at aztreonam concentrations of ≥ 64.0 $\mu\text{g/ml}$, while the 100-nm AgNPs decreased biofilm viability to $\sim 70\%$ at aztreonam

concentrations of ≥ 256 $\mu\text{g/ml}$ (Fig. 4B). With the 0.625- $\mu\text{g/ml}$ level of the 10-nm AgNPs, biofilms were nearly nonviable at all aztreonam concentrations tested (≥ 2.00 $\mu\text{g/ml}$) (Fig. 4D). As the concentrations of AgNPs were increased, the 20-nm AgNPs became more effective at reducing biofilm viability, followed by the 40-nm and 60-nm AgNPs (Fig. 4D to H). An exception to this general trend was the 100-nm AgNPs, which showed variable ef-

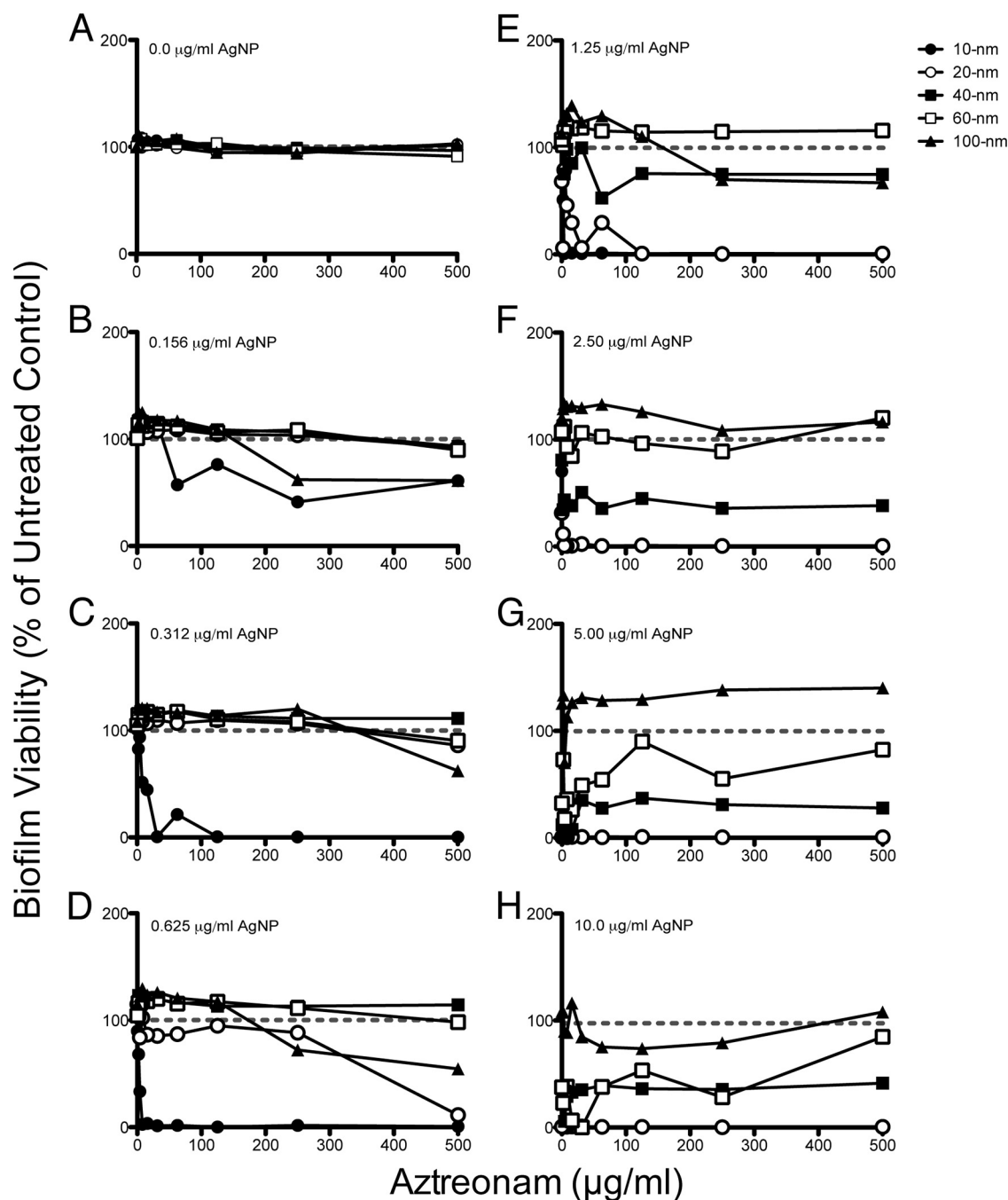


FIG 4 Effects of different AgNP-aztreonam combinations on *P. aeruginosa* PAO1 biofilm viability in the recovery phase. Normalized values of percent viability of biofilms following challenge with different combinations of aztreonam (0.0 $\mu\text{g/ml}$ to 512 $\mu\text{g/ml}$) and AgNP sizes (10 nm to 100 nm) at the specific AgNP concentration (0.0 $\mu\text{g/ml}$ to 10.0 $\mu\text{g/ml}$) indicated in the top left corner of each panel are shown (panel A represents replicate samples treated with aztreonam only). Biofilm viability values were normalized to values for untreated control samples (i.e., 0.0 $\mu\text{g/ml}$ AgNPs and 0.0 $\mu\text{g/ml}$ aztreonam). Dashed lines, 100% recovery (i.e., levels observed for untreated cells).

ffects on biofilm viability across both concentration gradients. It is unclear what factors may cause the variable inhibitory effects of the 100-nm AgNPs against biofilm viability, but it is clear, when these findings are combined with the corresponding biofilm biomass data for the 100-nm particles (Fig. 3), that there seems to be no overall negative effect on biofilm recovery. Together, these data suggest that, in combination with aztreonam, smaller AgNPs were

more effective in reducing *P. aeruginosa* PAO1 biofilm biomass and viability.

AgNPs and aztreonam exhibit antimicrobial synergy against *P. aeruginosa* biofilms. To directly compare the effects of combining AgNPs and aztreonam on biofilm biomass and viability, we used heat plots to visualize changes in these measures, expressed as percentages of untreated control values (Fig. 5). The heat plots

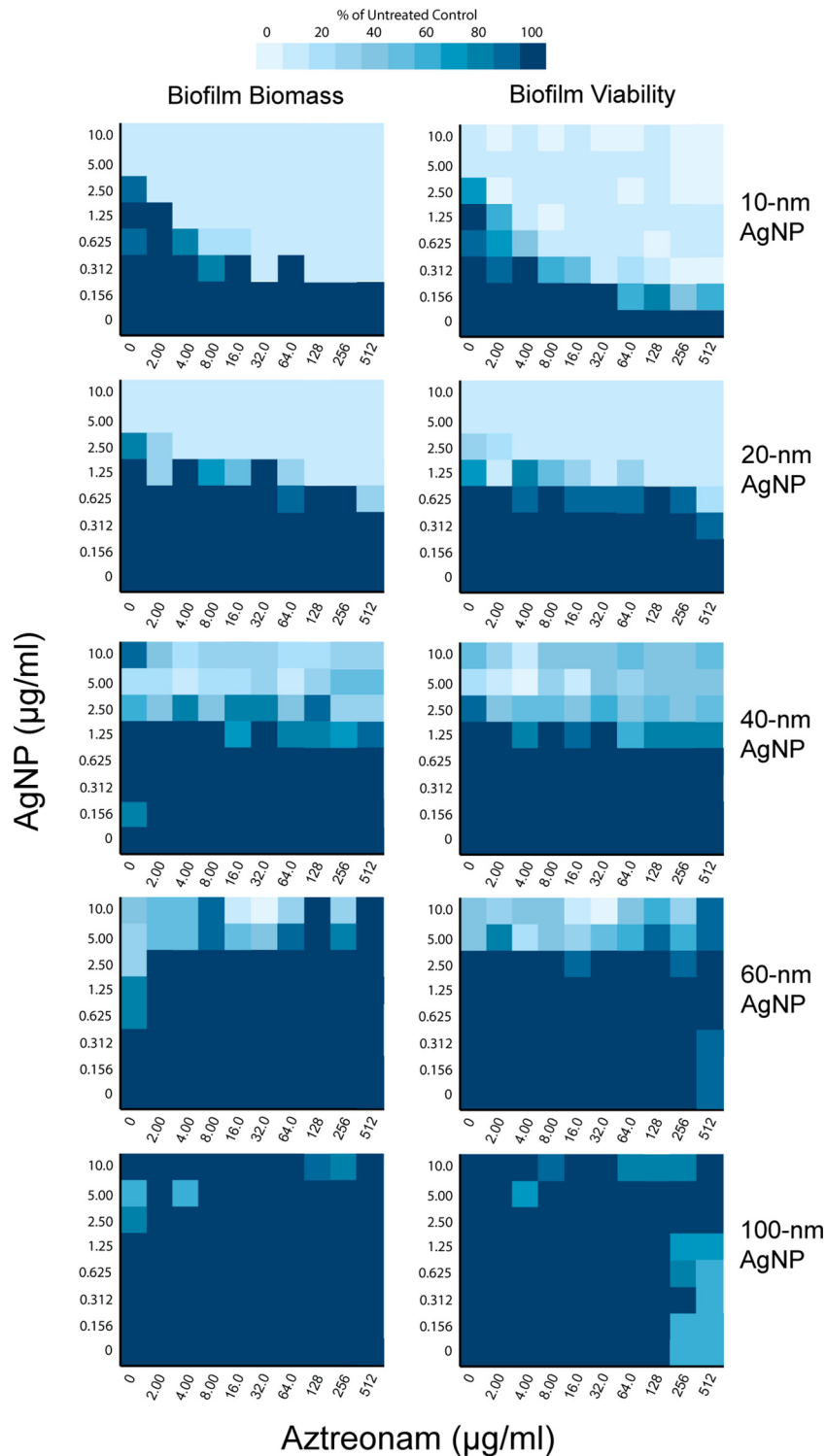


FIG 5 Comparison of effects of different AgNP-aztreonam combinations on biofilm biomass (left) and biofilm viability (right) in the recovery phase, according to AgNP size (indicated on the right). Results are presented as heat plots illustrating the normalized values of percent recovery of biofilm biomass and viability after treatment with AgNPs and aztreonam. One hundred percent recovery equals levels observed for untreated control samples. Darker blue boxes, greater percent recoveries (i.e., greater biofilm biomass or viability); lighter blue boxes, poorer biofilm recovery (i.e., greater antimicrobial effect).

indicated that smaller AgNPs had greater effects than larger AgNPs on biofilm biomass and viability at lower aztreonam concentrations. In addition, the data presented in Fig. 5 identified specific combinations of AgNPs and aztreonam concentrations

that may be synergistic in preventing recovery of *P. aeruginosa* biofilm populations following challenge.

To confirm and to quantify this potential synergy, we used the Bliss independence model and FIC index values to determine the

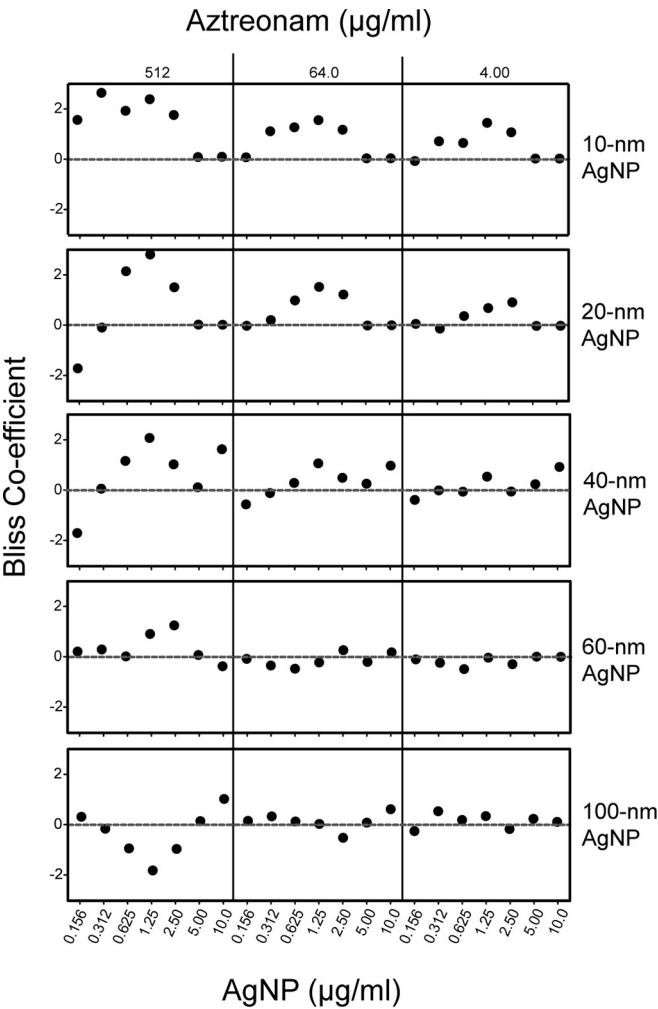


FIG 6 Bliss independence model of synergy, confirming synergistic antimicrobial effects of AgNP-aztreonam combinations against *P. aeruginosa* biofilms. The degree of synergy was quantified using the Bliss model of synergy for recovery based on the normalized values of percent biofilm biomass after 20 h of incubation in fresh medium. For each size of AgNPs (indicated on the right), each point on the graphs represents a specific combination of AgNPs (0.0 µg/ml to 10.0 µg/ml) and aztreonam (512 µg/ml, 64.0 µg/ml, or 4.00 µg/ml). Dashed lines, no combined effects of the antimicrobials. Positive Bliss coefficient values reflect synergy and negative values represent antagonism between the combined antimicrobials.

nature of the inhibitory effects observed when different antimicrobials were combined (35–38). We calculated the degree of antimicrobial synergy (Bliss coefficient) against *P. aeruginosa* biofilms for AgNPs of different sizes in combination with high (512 µg/ml), moderate (64.0 µg/ml), and low (4.00 µg/ml) concentrations of aztreonam (Fig. 6). In our calculations, positive values denote a synergistic combination of antimicrobials, and negative values describe antagonistic effects (37). The 10-nm AgNPs displayed the broadest profile of synergy over this range of AgNP and aztreonam concentrations (Fig. 6, top). The 20-nm and 40-nm AgNPs demonstrated a narrower synergy profile, with synergy revealed only at select AgNP and aztreonam concentrations. In contrast, the 60-nm and 100-nm AgNPs exhibited minimal synergy. Marked antagonism was even observed for the 100-nm AgNPs with 512 µg/ml aztreonam, with multiple values well below zero (Fig. 6, bottom left).

The evaluation of synergy using the FIC index indicated that the 10-nm and 20-nm AgNPs exhibited synergistic interactions with aztreonam, while the 40-nm, 60-nm, and 100-nm AgNPs showed no interactions (Table 3). Similar to the Bliss independence model data, the 10-nm AgNPs had the broadest interactions with aztreonam to inhibit the *P. aeruginosa* biofilms. While some 20-nm AgNP-aztreonam combinations did result in synergistic interactions, they were not as diverse as for the 10-nm AgNPs. This complemented our analysis using the Bliss model (Fig. 6). The primary differences occurred for the interactions of aztreonam with the 40-nm, 60-nm, and 100-nm AgNPs, with the FIC indices indicating no interaction and the Bliss independence model showing some interaction.

Visualization of the antimicrobial synergy of AgNPs and aztreonam against *P. aeruginosa* biofilms. To visualize the effects of AgNPs and aztreonam on *P. aeruginosa* biofilms, we used a flow reactor to culture biofilms inside silicone tubing (4, 33, 34) (Fig. 7A). Based on the calculations from the Bliss independence model (Fig. 6) and the MBEC values in Table 2, we selected (i) concentrations of aztreonam and AgNPs that were expected to be ineffective alone and (ii) concentrations of aztreonam and 10-nm AgNPs that displayed synergy against *P. aeruginosa* biofilms. Consequently, we treated the biofilms with 64.0 µg/ml aztreonam alone, 10-nm or 100-nm AgNPs alone (at 0.625 µg/ml), or combinations of these treatments, and we compared the thickness of the biofilms to that of untreated control biofilms (Fig. 7B). Full-field optical coherence tomography (FFOCT) results revealed that combinations of 10-nm or 100-nm AgNPs with aztreonam, and the 10-nm AgNPs alone, significantly reduced biofilm thickness versus untreated biofilms (Fig. 7B). The combination of AgNPs with aztreonam also significantly reduced biofilm thickness in comparison with either 10-nm or 100-nm AgNPs alone (Fig. 7B). Although both sizes of AgNPs plus aztreonam reduced biofilm thickness within the silicone tubing, the combination of 10-nm AgNPs and aztreonam was the most effective. These results support the role of the smaller AgNPs in promoting synergy, as illustrated in Fig. 6.

We next used transmission electron microscopy (TEM) to explore the physical changes that occurred at the cellular level upon exposure of *P. aeruginosa* cells within a biofilm to AgNP and aztreonam treatments. The same silicone tubing samples described above were embedded, thin sectioned, and imaged (45) to reveal

TABLE 3 Fractional inhibitory concentration index values calculated for AgNP-aztreonam combinations that inhibited *P. aeruginosa* PAO1 biofilms^a

Treatment	FIC index		Interaction
	Median	Range	
10-nm AgNPs	0.13	0.09–0.5	Synergistic
20-nm AgNPs	0.5	0.28–0.52	Synergistic
40-nm AgNPs	ND ^b	ND	None
60-nm AgNPs	ND	ND	None
100-nm AgNPs	ND	ND	None

^a Biofilm viability was assessed by measuring the OD₆₀₀ of the planktonic cultures in the wells of the microtiter plate following the 20 h of incubation of the recovery phase. ^b ND, not determined; none of the 40-nm, 60-nm, or 100-nm AgNP-aztreonam combinations prevented recovery of the biofilms after challenge. Some individual combinations did result in inhibition of *P. aeruginosa* PAO1 recovery (2 for 40-nm AgNPs, 1 for 60-nm AgNPs, and none for 100-nm AgNPs); however, these results appeared to be random.

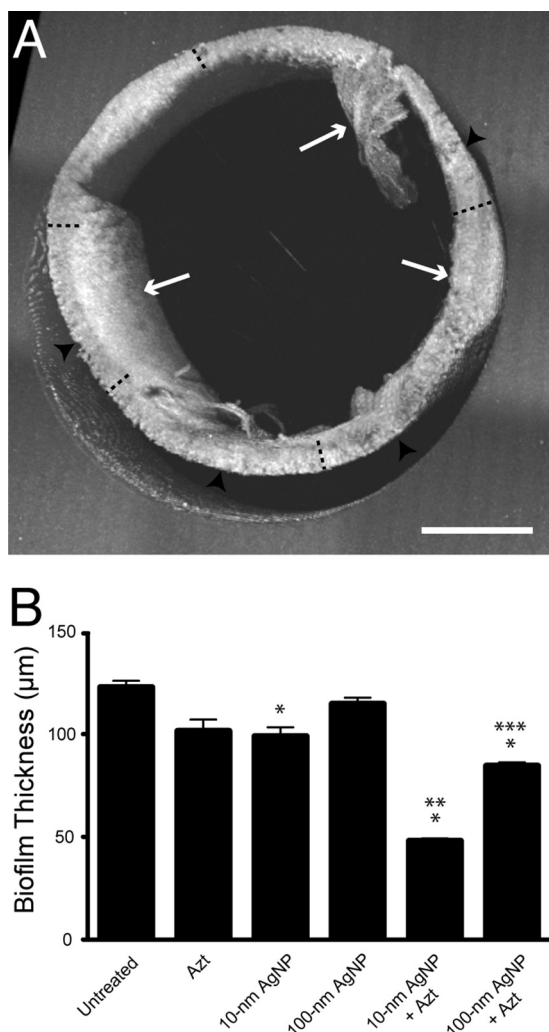


FIG 7 Effects of AgNP-aztreonam combinations on biofilm architecture. (A) Optical coherence tomographic slice of an untreated *P. aeruginosa* biofilm cultured for 72 h in silicone tubing. White arrows, biofilm; black arrowheads, areas where the biofilm detached from the inside of the silicone tubing. Black dashed lines, areas where measurements of biofilm thickness were derived. Bar, 400 μm. (B) Mean thickness (with standard deviation) of biofilms treated with various antimicrobials for 18 h ($n = 60$ measurements per condition). Antimicrobial treatments included 10-nm or 100-nm AgNPs at 0.625 μg/ml alone, aztreonam (Azt) at 64.0 μg/ml alone, and combinations of these treatments. *, $P < 0.05$ versus untreated control; **, $P < 0.05$ versus 10-nm AgNPs alone; ***, $P < 0.05$ versus 100-nm AgNPs alone.

the morphology and ultrastructure of cells and the internal structure of the biofilms (Fig. 8). Micrographs of untreated biofilm controls (Fig. 8A) showed the presence of intact *P. aeruginosa* cells and biofilm matrix material. Biofilms treated with 64.0 μg/ml aztreonam alone or either 10-nm or 100-nm AgNPs alone (Fig. 8B, C, and D, respectively) showed no significant morphological or ultrastructural defects in comparison with untreated cells (Fig. 8A). However, biofilms treated with a combination of 10-nm AgNPs and aztreonam exhibited substantial cellular changes, including altered cellular morphology and irregularities in both the distribution of the membrane and the cytoplasmic contents of the cells, indicating that the integrity of the bacterial cell walls was likely compromised (Fig. 8E). Furthermore, 10-nm AgNPs could

be detected throughout the matrix of these biofilms and were associated with cellular debris (Fig. 8G). In contrast, biofilms treated with 100-nm AgNPs and aztreonam exhibited minor cellular perturbations, compared to cells treated with 10-nm AgNPs plus aztreonam, and no 100-nm AgNPs were observed within the biofilm matrix posttreatment (Fig. 8F).

DISCUSSION

Chronic infections mediated by bacterial biofilms are often highly resistant to conventional antibiotic treatments. This is due, in part, to the inherent architecture of the biofilm that encases and protects bacterial cells in a matrix containing exopolysaccharides, DNA, outer membrane vesicles, and other cell-derived content (2, 3, 33, 46). These components of the biofilm matrix have been proposed to absorb and even sequester antibiotics away from the viable cells contained within the matrix (2, 5). In chronic *P. aeruginosa* infections, this antimicrobial resistance is further heightened by other adaptive mechanisms and the virulence factors that enhance the organism's ability to resist conventional antibiotic treatments (3, 10, 11).

In this study, we assessed the efficacy of citrate-capped AgNPs of different sizes, alone and in combination with the antibiotic aztreonam, to prevent the recovery of *P. aeruginosa* PAO1 biofilms *in vitro*. Our results indicated that, when used alone, biofilm biomass was actually enhanced with aztreonam treatment. Even at concentrations as high as 512 μg/ml aztreonam, the treatment had no lasting effect on the viability of cells within the biofilm (Fig. 2A and Table 2). Similar effects have been observed for another β-lactam antibiotic, with exposure of *P. aeruginosa* PAO1 biofilms to sub-MIC levels of imipenem increasing biofilm biomass via elevated alginate production (47). In our experiments, biofilms were exposed to aztreonam at concentrations up to several hundred times the MIC but recovered after challenge, with significant increases in biomass (Fig. 2A). These results are also consistent with previously published reports of wild-type *P. aeruginosa* K767 exhibiting MBEC values of $>1,000$ μg/ml against aztreonam (48). Furthermore, visualization of biofilms in this study revealed that treatment with aztreonam alone had virtually no effects on biofilm architecture, cellular morphology, or the ultrastructure of cells within the biofilms (Fig. 7B and 8B). This suggests that the *in vitro* inhibitory effects of this antibiotic against preformed biofilms are limited and biofilm growth may even be enhanced after antibiotic treatments have ceased. Aztreonam alone did prevent the growth of planktonic cells that emerged from the biofilm during the challenge phase of the experiment, at concentrations of ≥ 2.00 μg/ml (Table 1), which is consistent with the reported MIC for aztreonam against planktonically grown *P. aeruginosa* of 2 to 4 μg/ml (49).

In contrast, the smaller 10-nm or 20-nm AgNPs assessed individually were more effective than aztreonam alone in preventing the recovery of both biofilm and planktonic cells in the MBEC assays (Fig. 2B and C). However, the 10-nm AgNPs alone presented only minor, albeit significant, defects in biofilm architecture, as visualized by FFOCT (Fig. 7B), and no obvious alterations in cellular morphology or ultrastructure, as visualized by TEM (Fig. 8C). In contrast, the 40-nm and 60-nm AgNPs demonstrated limited inhibition of biofilm biomass and viability in the MBEC assays, while the 100-nm AgNPs showed no major inhibitory effects in any of the assays (Fig. 2B and C and 7). In comparison, in the presence of all sizes of AgNPs tested, we observed inhibition of

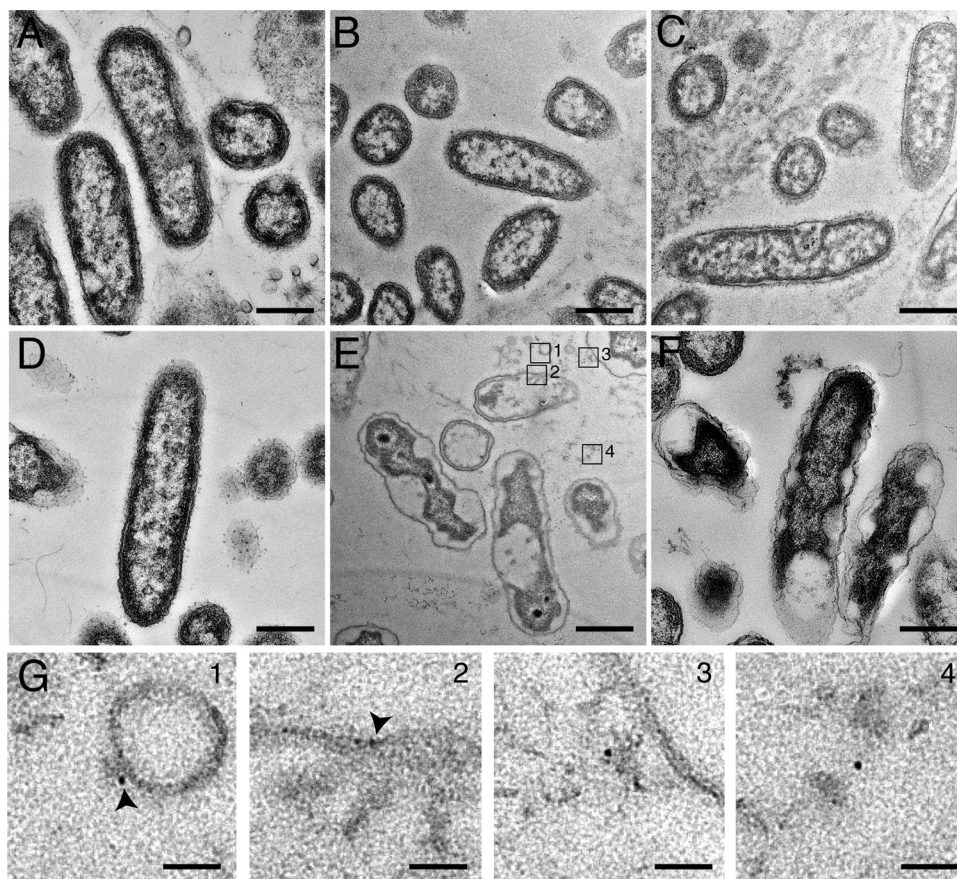


FIG 8 Penetration of smaller AgNPs into *P. aeruginosa* biofilms. (A to F) Negatively stained transmission electron micrographs of *P. aeruginosa* PAO1 biofilms cultured in silicone tubing for 72 h and either left untreated (A) or treated for 18 h with aztreonam (64.0 $\mu\text{g/ml}$) (B), 10-nm AgNPs (0.625 $\mu\text{g/ml}$) (C), 100-nm AgNPs (0.625 $\mu\text{g/ml}$) (D), 10-nm AgNPs (0.625 $\mu\text{g/ml}$) plus aztreonam (64.0 $\mu\text{g/ml}$) (E), or 100-nm AgNPs (0.625 $\mu\text{g/ml}$) plus aztreonam (64.0 $\mu\text{g/ml}$) (F). Bars, 500 nm. (G) Inset images from panel E, highlighting the presence of 10-nm AgNPs (arrowheads). Bars, 50 nm.

planktonic cells during the challenge phase of the experiment, with median MIC values ranging from 0.234 $\mu\text{g/ml}$ for 10-nm AgNPs to 7.50 $\mu\text{g/ml}$ for 100-nm AgNPs (Table 1). These findings suggest that the AgNPs themselves possess finite antimicrobial capabilities against biofilms, and they are consistent with the resistance of biofilms to concentrations of antimicrobials that effectively inhibit their free-living counterparts.

Several studies have reported antimicrobial activities of AgNPs against a wide range of planktonic Gram-positive and Gram-negative bacterial cultures, with similar MIC values (50, 51), although many of those studies did not evaluate variations in efficacy based on the size of the AgNPs. A recent study by Ivask et al. investigated AgNPs of various sizes and surface charges for their toxicity against planktonic *Escherichia coli* cells (52). Their results suggested that smaller AgNPs resulted in greater silver ion dissolution and toxicity against *E. coli* cells (52). That study also concluded that the physiochemical properties of the AgNPs, particularly the surface characteristics, were significant factors in triggering specific bacterial responses (52). Those bacterial responses suggest the action of reactive oxygen species as the mechanism of action, as a consequence of the dissolution of silver ions from the nanoparticles (22). A study by Loo et al. examined the detachment of preformed *P. aeruginosa* biofilms treated with AgNPs of different sizes (7 nm to 70 nm; synthesized using an ascorbic acid/citrate

seed-mediated method) and visualized by confocal laser scanning microscopy (53). The authors found that concentrated AgNP solutions (600 $\mu\text{g/ml}$) with average diameters between 8 nm and 10 nm were most effective and reduced *P. aeruginosa* biofilms by >90% (53).

Building on our initial results, we added an antibiotic commonly used to treat CF lung infections to our MBEC assays, and we observed the greatest reductions in biofilm biomass and viability by combining AgNPs with aztreonam (Fig. 3 and 4). Once again, 10-nm and 20-nm AgNPs demonstrated more-significant reductions in biofilm recovery when combined with lower concentrations of aztreonam than did 40-nm and 60-nm nanoparticles. Next, we used the Bliss independence model (22, 37, 54) and FIC index (24, 39, 40) to calculate synergy between AgNPs and aztreonam in the inhibition of *P. aeruginosa* biofilm biomass (Fig. 6 and Table 3), and we confirmed that 10 nm was the optimal AgNP size to use in the antimicrobial combination. Interestingly, while the Bliss independence model indicated that some measurable levels of synergy were observed for all AgNPs tested, the FIC indices indicated synergy only for the 10-nm and 20-nm AgNPs. However, the 10-nm AgNPs showed the highest degree of synergy even at the lowest concentrations of aztreonam and AgNPs with both methods of calculating synergy. This is important for several reasons: (i) the known cytotoxic effects of AgNPs (reviewed in

reference 14) can be minimized only by reducing the concentration used to treat the infection, (ii) smaller nanoparticles should have greater utility in *in vivo* environments such as the lung, where smaller bronchial passages can become colonized by *P. aeruginosa* (11, 55), and (iii) decreasing the required concentration for an antibiotic to function successfully against biofilm-bound microorganisms will help minimize the risks associated with developing further antibiotic resistance due to the protective effects of biofilms. With the difficulty of accumulating sufficient concentrations of antibiotics in the lungs to effectively eradicate biofilms, the risk for development of resistant strains increases to the detriment of the patient.

A recent study by Markowska et al. described synergistic inhibitory effects against *P. aeruginosa* ATCC 10145 of mixtures of uncapped AgNPs of different sizes (2 nm to 35 nm) and a range of antibiotics (24). The authors found that their AgNP-antibiotic combinations were highly effective against planktonic cells, but they observed no synergy against *P. aeruginosa* biofilms. The authors concluded that the inability of the uncapped AgNP-antibiotic combinations to effectively inhibit biofilms was primarily due to a lack of antimicrobial exposure. They argued that this lack of exposure was caused by both the inherent protection of the extracellular matrix that accompanies *P. aeruginosa* biofilm development and aggregation of the uncapped AgNPs, which led to an increase in size and therefore a decrease in ability to penetrate the biofilms (24). A recent study by Morones-Ramirez et al. described the effects of silver ions and their synergy with antibiotics to treat a range of infections both *in vivo* and *in vitro* (22). Their results demonstrated that silver ion-antibiotic combinations increased the production of reactive oxygen species, which in turn increased the membrane permeability of Gram-negative bacteria. The authors hypothesized this may potentiate the activity of a broad range of antibiotics against Gram-negative bacteria in different metabolic states, as well as restoring antibiotic susceptibility to resistant bacterial strains. This synergistic effect and the results for the uncapped AgNPs reported by Markowska et al. (24), as described above, are consistent with the findings we observed through imaging of preformed *P. aeruginosa* biofilms. In our study, the selected combination of 10-nm citrate-capped AgNPs and aztreonam caused significant defects in biofilm architecture, membrane permeability, and ultrastructure, in comparison with all other antimicrobial treatments evaluated (Fig. 7 and 8). These data demonstrate the ability of small citrate-capped AgNPs to synergistically enhance the antimicrobial effects of aztreonam against *P. aeruginosa* biofilms *in vitro*, and they support a potential role for these small nanoparticles in reducing the antibiotic burden of patients treated for chronic infections.

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